CLASS 8 — SCIENCE · HARD LEVEL UNIT TEST SET 2

Chapters 5–9: **Exploring Forces, Pressure, Winds, Storms and Cyclones, Particulate Nature of Matter, Nature of Matter: Elements, Compounds and Mixtures, The Amazing World of Solutes, Solvents and Solutions** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Assertion (A): A sky-diver falls at constant speed after opening the parachute. Reason (R): The forces on her are balanced. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. Assertion (A): Dams are made thicker at the bottom. Reason (R): Water pressure increases with depth. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

3. Assertion (A): A gas can be compressed easily but a solid cannot. Reason (R): Gas particles are widely separated while solid particles are tightly packed. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

4. Assertion (A): Brass is a mixture of copper and zinc. Reason (R): The proportions of copper and zinc in brass can vary. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

5. Assertion (A): A soft drink goes flat when left open. Reason (R): The solubility of a gas in a liquid decreases as pressure decreases. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

6. A man weighs 600 N on Earth. On a planet where gravity is 1/3 of Earth's, his weight is: 1

(a) 300 N **(b)** 600 N **(c)** 1800 N **(d)** 200 N

7. A box exerts 2000 Pa. If its base area is halved without changing weight, the pressure becomes: 1

(a) 500 Pa **(b)** 1000 Pa **(c)** 4000 Pa **(d)** 2000 Pa

8. A sample of a pure solid is heated at a constant rate and the temperature is recorded. A horizontal portion on the graph shows: 1

(a) boiling only **(b)** change of state **(c)** chemical change **(d)** freezing only

9. 9 g of water is decomposed. The masses of hydrogen and oxygen obtained are: 1

(a) 4.5 g and 4.5 g **(b)** 8 g and 1 g **(c)** 1 g and 8 g **(d)** 2 g and 7 g

10. A solution is made by mixing 200 g of 10 per cent solution and 300 g of 20 per cent solution of the same solute. The percentage of the final solution is: 1

(a) 14 per cent **(b)** 15 per cent **(c)** 16 per cent **(d)** 30 per cent

11. A magnet is brought near an iron nail and the nail jumps to the magnet. The force that does the work is: 1

(a) electrostatic force **(b)** friction
(c) gravity **(d)** magnetic force

12. A ship at sea is advised to move away from the path of a cyclone because: 1

(a) the eye is cold
(b) the sea is calm
(c) wind and waves are strongest near the eye wall
(d) ships are heavy

13. **Assertion (A):** Oxygen is an element. 1
Reason (R): Oxygen contains only one kind of atom and cannot be broken down into simpler substances by chemical methods.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

14. **Assertion (A):** A solution of salt in water does not scatter a beam of light. 1
Reason (R): The particles in a true solution are extremely small.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

15. **Assertion (A):** The mass of an object is the same on the Earth and on the Moon. 1
Reason (R): Mass is the amount of matter in an object, which does not change with place.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

MASTERMIND

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Why does a block slide down a slope faster when the slope is polished? 2

17. State why weather satellites are used for cyclone tracking. 2

18. Why does sweating not cool us effectively when the humidity is high? 2

19. Explain why alloys like solder are used to join metals. 2

20. Explain why boiled and cooled water tastes flat. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Compare ice, water and steam in a table in terms of energy, spacing and motion of particles. 3

22. Explain why diamonds do not conduct electricity but graphite does. 3

23. Explain how crystallisation purifies copper sulphate. 3

24. Why does an object dropped from a height of 1 m take less time to reach the ground than from 2 m? 3

OR

Explain why helium balloons rise but ordinary air balloons do not. 3

25. How would the boiling point change if you added salt to water? Explain briefly. 3

OR

A fruit juice is made of water, sugar, acids and colour. Describe its nature and how you may separate water from it. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. **Case study.** Mohan weighs 600 N. When he stands on both feet, each foot has an area of 0.01 square metre in contact with the ground. When he lifts one foot, he stands on one foot only. 5

(i) State the formula for pressure.
(ii) Find the pressure when he stands on both feet.
(iii) Find the pressure when he stands on one foot.
(iv) A camel's wide feet do not sink in the desert sand, but a person's narrow heels do. Explain.

27. A cylinder with a piston traps air. Explain what happens to the pressure when the piston is pushed in or pulled out, and relate it to a bicycle pump. 5

OR

Calculate: how many times does the volume increase when 1 mL of water changes to steam if the volume of steam is about 1600 mL at 100 degrees Celsius? Explain the reason. 5

END OF QUESTION PAPER — Total: 50 Marks

INTERNAL